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Design for automated disassembly: a comparative study of different battery component designs

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ABSTRACT

The feasibility of automated disassembly at a product's end-of-life stage strongly depends on its design. This relationship is particularly relevant for electric vehicle batteries, for which other design requirements are typically given priority. Consequently, the potential for automated disassembly varies between different battery designs and between the different components. This study investigates the potential for automated disassembly of five EV battery designs currently available on the market. These batteries are subject to disassembly experiments, during which a criteria catalogue is employed to semi-quantitatively compare the individual design characteristics of each individual component. The results of this assessment indicate that the complexity involved in automatically disassembling the components generally increases with the depth of disassembly. Nevertheless, the initial step of the system cover removal appears to be of high importance. It is further identified that the steps involved in the removal of cooling systems, the separation of module housing parts and the cell connector separation are the most critical during disassembly. Based on these findings specific design guidelines are proposed from the best evaluation results of the five disassembly analyses performed.

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Lithium-ion battery; automation; design for recycling; design for disassembly; circular economy; end of life; comparative assessement

1. Introduction and motivation

Electric vehicles (EVs) powered by lithium-ion batteries (LIBs) play a central role in transforming the transport sector towards climate neutrality by gradually replacing internal combustion engine vehicles. It is anticipated that the annual demand for LIBs in Europe will increase from 60 GWh in 2020 to 1,068 GWh in 2040. Of this projected growth approximately 800 GWh is expected to be utilised in EVs (Neef, Schmaltz, and Thielmann 2021). Consequently, the annual return quantities of these batteries are expected to reach between 0.2 and 1.5 mt by 2040 (Neef, Schmaltz, and Thielmann 2021; Thompson et al. 2020).

In general, end-of-life (EoL) battery packs from EVs are initially disassembled manually to a certain extent before the components are assigned to specific material recycling routes. Battery technology is still undergoing significant advances. This includes for example process technology advances such as dry coating the application of new cell chemistries and innovations regarding all-solid-state batteries. As a consequence, a rather heterogeneous representation of EV battery designs is currently available on the market, which is a trend that can be expected to continue in future (Gaines, Richa, and Spangenberger 2018). Previous studies found that these different designs often show very large differences in terms of their individual facilitation or hindrance of disassembly (Lander et al. 2023; Ohnemüller et al. 2025; Thompson et al. 2020). Most designs only partially enable efficient dismantling, necessitating the involvement of manual labour to a considerable extent. So far, however, there has been no comprehensive investigation on specific battery component designs and whether they promote or impede automated disassembly processing.

In this context, a comparative analysis of five battery designs from different manufacturers is presented in this study. These designs are structurally investigated applying a methodology that takes these insights as a basis to eventually provide component-specific design guidelines that promote automated disassembly. These guidelines can be integrated into a product development process to shape the design process of batteries to promote their automated disassembly. The main contributions of this work are the following:

(1) Compilation of relevant automation requirements for EV batteries,
(2) Detailed comparison of market-available battery designs regarding their potential to enable automated disassembly at the component level,

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(3) Provision and application of a methodology for the derivation of guidelines for the (re-)design allowing disassembly automation.

Therefore, in this study we present a practical methodology that enables the inclusion of battery disassembly requirements in product development, focusing on automated processes and resulting in according battery design guidelines. The following sub sections explain a typical battery disassembly procedure and according design issues as well as different design principles. Further, the current state of research is examined. Section 0 outlines the proposed systematic methodology, detailing the steps from disassembly analyses to the derivation of design guidelines. Section 3 presents the application of this methodology and its results, visualisation of disassembly processes, and identification of components suited for automation. The study closes with a conclusion also explaining its limitations.

1.1. Typical battery disassembly process

The 9 R framework according to the United Nations Environment Program (UNEP) structurally presents circular economy strategies in a hierarchical order (United Nations Environment Programme 2019). This hierarchy is based on the potential for the reduction of a products' environmental impact by value retention and is divided into three main categories: immaterial, product-level, and material-level strategies. Immaterial strategies aim to reduce or eliminate the need for physical products altogether, while product strategies extend the life and value of existing products through maintenance or reconfiguration. Material strategies, like recycling, come last in the hierarchy as they typically require more energy and result in lower value recovery. The disassembly of traction batteries is a crucial step in preparing for most of these 9 R strategies. Especially strategies that enable a high value retention by extending the lifetime like reuse (R3) and repair (R4) depend on a certain level of disassembly. But also, the sheer recovery of material by recycling processes (R8) can benefit from an early separation of materials that is enabled by disassembly. This is because the prior removal of components leads to smaller material quantities and pure material flows to be treated in the subsequent process chain (Guo et al. 2021).

Many current battery designs primarily use between two and four different joining techniques, only some of which can be disassembled in a non-destructive manner (Thompson et al. 2020). Typically, screws, nuts and bolts, rivets and firmly bonded joining techniques like welding and gluing are applied for building the module housing and encasing the cells (Das et al. 2018; Wessel et al. 2021). In general, bolted joints facilitate the disassembly, whereas adhesive joints are often difficult or impossible to disassemble in a non-destructive manner (Scott et al. 2023). Non-destructive disassembly, however, is an important prerequisite for product-level R-strategies that enable further life phases (refurbish, R6 and remanufacturing, R7). The optimal battery disassembly procedure is highly specific to the individual design and the economic and environmental targets. Nevertheless, some steps can be generalised or summarised at a higher aggregation level. A typical disassembly process chain is shown in Figure 1 (Zorn et al. 2022).

The general steps for a high disassembly depth essentially comprise the opening of the battery pack, the removal of the

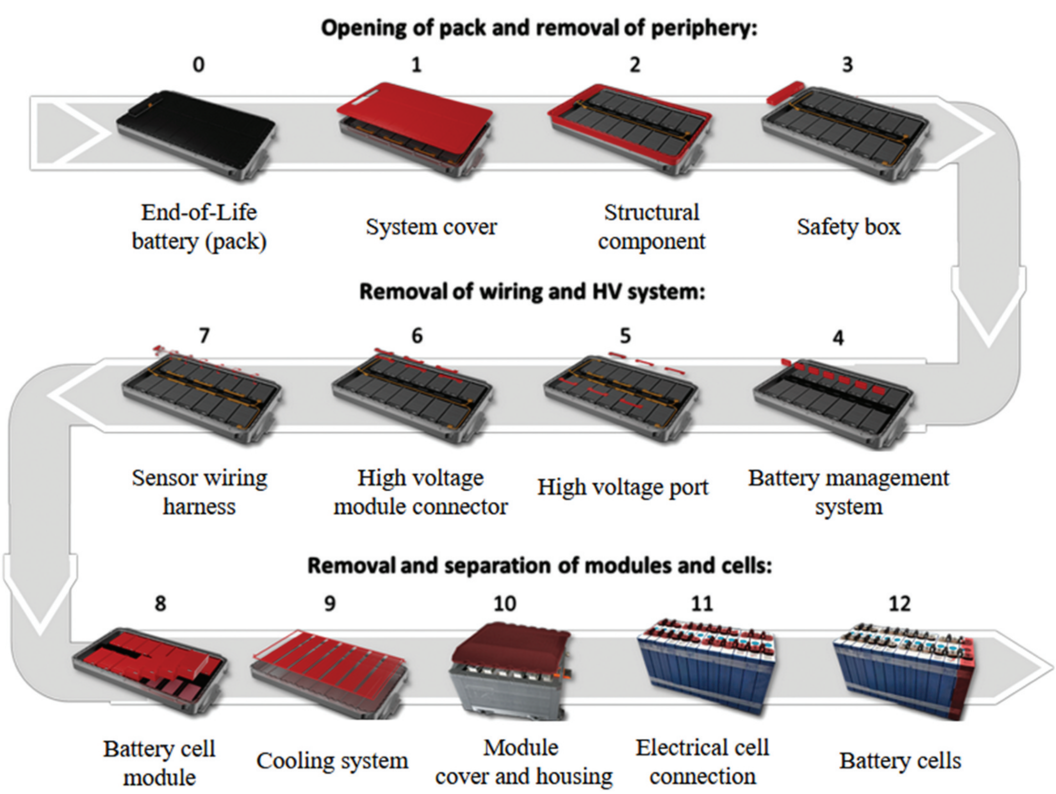


Figure 1. Standard disassembly procedure of a battery with standard pack – module – cell design. Disassembly depth increases from 1–12; adapted from (Zorn et al. 2022).

cover and periphery (1–3), the removal of wiring and the high voltage (HV) as well as battery management systems (BMS) (4–7) and finally the removal and separation of modules and cells (8–12). One approach to simplify this process chain would be the purposeful design of the battery to consider the specific requirements these processes have, typically summarised under the term Design for X (DfX).

1.2. Motivation for design for X and Methodologies for structured product development

To enable a cost efficient treatment of the fast rising number of EoL batteries, the employment of automated processing will be necessary (Villagrossi and Dinon 2023).

It has already been shown that automation in EoL processes can be economically beneficial compared to manual disassembly (Görgens et al. 2025). However, at present automation in battery EoL treatment is only applied in exceptional cases. This can be attributed to three principal challenges:

(1) Batteries are individually designed for different electric vehicles (OEM and segment-specific), which results in a lack of uniformity in terms of structure, composition and joining techniques (Blankemeyer et al. 2021; Zorn et al. 2022).
(2) Detailed information regarding the product and its state are essential for the targeted dismantling process. However, this information is often not accessible to the dismantlers (Zorn et al. 2022).
(3) The current focus of battery design is on the facilitation of its manufacturing and the improvement of battery characteristics during the use phase (Harper et al. 2023; Rönkkö et al. 2024; Thompson et al. 2020). As a result, joining techniques such as gluing or welding are frequently employed, which renders disassembly challenging (Scott et al. 2023). Furthermore, the design of battery components does not thoroughly consider non-destructive disassembly nor other requirements such as accessibility or enabling the use of common tools (Thompson et al. 2020).

As a result of these three issues, a high depth of disassembly often becomes cost-prohibitive due to high labour cost for manual disassembly and the process often has to be terminated prematurely (Lander et al. 2023; P. Li et al. 2024). Consequently, numerous materials remain in the recycling stream, which could have been removed at an earlier stage. This results in the accumulation of larger material volumes with higher material mixing in downstream recycling processing chains, which increases the effort required for an effective separation (Santini et al. 2010; Zorn, Hams, and Flamme 2024). Accordingly, the energy demand increases, which has a negative effect on the environmental impact and imposes substantial cost burdens on companies that are active in performing EoL operations. This issue is of particular concern considering the anticipated increase in product return volumes described above. Furthermore, an efficient and automated disassembly process could also facilitate the increased application of reuse strategies (Baazouzi et al. 2021). These often require a higher disassembly depth as well as a faster condition assessment than can currently be achieved with the battery designs available on the market.

In this context, a new Battery Regulation has been introduced in the European Union in 2023 to replace the previous Battery Directive (European Union 2023). In future, recyclers will have access to disassembly information for batteries as well as data on the battery structure, composition and the battery chemistry. This will be achieved by means of a battery passport (European Union 2023). The information provided could serve as a key enabler of more sophisticated disassembly strategies, as it addresses the prevailing information gaps (Hansen et al. 2024). It can be assumed that this will make a significant contribution to solving the first of the challenges listed above. The second and third challenge, however, remain to be dealt with. It is unlikely that there will be a comprehensive standardisation of the EV battery system design across different OEMs (Sanguesa et al. 2021).

Design for X is a principle that encourages designing products or systems with a specific focus or intention that is represented by the X. This can be life cycle stages such as assembly or manufacturing or specific product performance criteria like reliability, safety or the environment (Pahl et al. 2007). The goal is to optimise the design by specifically considering specific perspectives and requirements, that might otherwise not or only to a limited extent be included. The concept of Design for Disassembly (DfD) has gained attention since it enables reuse and repurpose practices and thus elongates the lifetime of products (Bogue and Lowe 2007).

Methods to improve a product's dismantlability through DfD include the purposeful selection and use of materials, the design of components and the product architecture, as well as the optimisation of joint types (Bogue and Lowe 2007; Kroll and Hanft 1998). Especially in the case of LIBs, the design for disassembly should also consider the specific requirements of automation in EoL processing. Due to the expected return volumes in the future, their automated processing will be a prerequisite for an economically feasible EoL treatment. Therefore, the focus is on 'Design for Automated Disassembly' (DfAD).

To aid product development, the DfAD guidelines developed must integrate seamlessly into existing product development procedures such as the waterfall, agile methodology, or the V-model, which all start with deriving product requirements. In the case of the V-model, requirements are based on stakeholder expectations, that are reviewed by the application of checkpoints to ensure comprehensiveness and resolve contradictions. Although recycling in general is considered during these checkpoints, automated disassembly is not. Thus, it would be beneficial to include remanufacturing and recycling companies as stakeholders to derive specific design requirements for automated disassembly, especially for battery components.

1.3. Related research

There have been several studies on the evaluation of the dismantlability for different product designs in general as well as for batteries in particular. Some of these studies rely on the application of an attribute or criteria list, to generate quantifiable values based on the respective level of their fulfilment. For the purposes of categorisation, these are compared with the scope of the present work. Accordingly, the criteria by which a relevant publication is considered to be related to the contents of this work are as follows:

I	Comparison	Does the study compare different design possibilities?
II	Component design	Does the study specifically consider component design level?
III	Requirements list	Is a structured and comprehensible criteria list employed to evaluate the designs?
IV	Automated disassembly	Does the study specifically consider automated disassembly? (in contrast to manual disassembly only)
V	Derived guidelines	Does the study specifically provide guidelines to help improve designs?
VI	Application case: Batteries	Are EV batteries the application case of the study?
VII	Year	Has the study been published recently?
VII	Product development	Does the study specifically consider the product development?
IX	Disassembly performed	Are purposeful disassembly analyses conducted to support the evaluation?
X	Product centered	Are EoL-processes or products evaluated in the framework of the study?

A study is considered appropriate to represent the state of research if it generally addresses the topic of evaluating the suitability of different designs for disassembly and if it additionally addresses several of the criteria listed above. Table 1 provides an overview of the identified literature in the context of this publication.

Gschwendtner and Kopacek (1996) introduce the autodemograph method for assessing product designs for automated disassembly and recycling. The method applies a data structure designed to present essential information and includes the use of comparative indices. In a similar sense, Kroll and Carver (1999) propose a method that is able to estimate design-specific manual disassembly times based on four distinctive criteria. These are the accessibility and positioning, the force required and a base time criterion which is used as a general basis for estimating the time required to disassemble a certain type of joint. Based on an existing approach, Herrmann and Raatz et al. (2012), developed a catalogue of criteria to assess whether and to what extend the automation of a certain process is technically feasible. The authors apply this catalogue to the disassembly procedure of an existing battery and find the specific automation potentials for each disassembly step. A similar catalogue is applied in an adapted form by (Hellmuth, DiFilippo, and Jouaneh 2021), who investigate the disassembly processes for more recent battery designs and apply a more detailed quantification method. (Desai and Mital 2003) introduce a comprehensive methodology aimed at improving dismantlability through a systematic integration of disassembly

Table 1. Highlighting of main gaps in literature, based on relevant criteria for the assessment of component-level design for automated disassembly capabilities of different designs. The greater the degree of colour filling of the circle the higher the level of fulfilment of the respective criterion. The translation of the different circles can be found in the appendix. The translation of the symbols can be found in appendix 1.

Author(s) (year)	I. Comparison	II. Component design	III. Requirements list	IV. Automated disassembly	V. Guidelines derived	VI. Application case: Batteries	VII. Year	VIII. Product development	IX. Method: disassembly	X. Product design centered
Gschwendtner and Kopacek (1996)	○	◑	○	●	○	○	○	●	○	●
Kroll and Carver (1999)	○	●	●	○	○	○	○	◔	○	●
Desai and Mital (2003)	○	●	●	○	●	○	○	●	○	●
Sodhi, Sonnenberg, and Das (2004)	○	●	●	○	○	○	○	○	●	●
Tseng et al. (2008)	○	●	●	○	○	○	◔	◑	○	●
Herrmann et al. (2012)	○	○	●	●	○	●	◑	○	●	○
Weyrich and Natkunarajah (2013)	○	○	○	●	●	●	◑	○	○	○
Soh, Ong, and Nee (2014)	○	◑	○	◑	◔	○	◕	◑	○	○
Aguiar (2016)	○	●	●	○	◕	●	◕	◑	●	●
Li (2019)	○	●	●	●	○	○	◕	○	○	○
Blankemeyer et al. (2021)	●	◔	◔	◕	◔	●	◕	○	◑	○
Hellmuth, DiFilippo, and Jouaneh (2021)	◑	○	●	●	○	●	◕	○	●	○
Rosenberg et al. (2022)	○	◑	○	○	◑	●	●	◑	●	●
Lander et al. (2023)	●	◑	○	◔	○	●	●	◔	◔	●
Baazouzi, Grimm, and Birke (2023)	○	◔	○	◑	○	●	●	◑	●	●
Sierra-Fontalvo et al. (2024)	◑	●	●	○	○	○	●	◑	●	●
This study	●	●	●	●	●	●	●	◕	◕	●

considerations into product design. They facilitate quantitative evaluation by assigning numerical indices to design factors, thereby helping to identify disassembly challenges. Weyrich and Natkunarajah (Weyrich and Natkunarajah 2013) evaluate various systems for identifying the battery components and techniques for loosening the joints. From this, the authors developed a semi-automated disassembly approach using modular components with the possibility of establishing a fully automatic process. Different DfD methods and according disassembly procedure of products are discussed by Soh et al. (Soh, Ong, and Nee 2014). By considering disassembly constraints and requirements early in the design phase, products can be optimised for ease of disassembly, material recovery, and reuse of components and sub-assemblies. Sodhi et al. (Sodhi, Sonnenberg, and Das 2004) determine the unfastening effort by the introduction of an unfastening model through which the number of unfastening actions and the properties of the connecting machine elements are determined. Correction factors are then assigned to the respective properties such as thread length or wrench width and these are summed up to quantitatively describe the total effort for disassembly. Tseng et al. (Tseng, Chang, and Li 2008) use the modelling of a liaison model as an approach to describe products in terms of their structure and to evaluate the dependencies by determining the liaison intensity. Using grouping genetic algorithms and the subsequent analysis of the materials used, an approach is presented to improve the life cycle characteristics of the initial product. Li et al. (J. Li, Barwood, and Rahimifard 2019) investigate the trade-offs between environmental benefits, technical feasibility and economic viability of robotic disassembly in EoL processes. To do this, they develop a decision support tool that employs a wide range of different assessment indicators that represent the different sustainability perspectives. Detailed product design considerations regarding their automation possibilities in disassembly are considered by Blankemeyer et al. (Blankemeyer et al. 2021). In their study, the authors classify the joint types present in LIBs and support the design planning of an automated disassembly system based on the prior product analysis. Their study concludes that a certain degree of uniformity in battery design would be beneficial but remains a challenge in the future. Lander et al. (Lander et al. 2023) investigate the difference in disassembly costs between five commercially available battery pack designs. In their analysis they apply a combined techno-economic perspective to investigate the economic consequences of increased disassembly times. They apply joint type-specific time demand values that corresponding removal tasks would have and calculate the total time required for the different designs. They conclude that disassembly times are highly design-dependent and that at least partial automation could already result in a significant reduction of disassembly costs. Sierra-Fontalvo et al. (Sierra-Fontalvo et al. 2024) introduce a diagnosis tool that identifies a product's remanufacturing potential at component level. They address the design-related dismantlability of the product and also consider its physical condition upon return. The first is calculated based on a dismantlability index. This index evaluates the type of joint to be unfastened, its accessibility and complexity as well as its disassembly requirements, which include the necessity of specialised tools. Baazouzi et al. (Baazouzi, Grimm, and Birke 2023) developed a design for an adaptive disassembly planner. The tool is evaluated on a specific battery design (used e.g. in Audi A3 Sportback Hybrid). Three decisions are combined: the optimal disassembly sequence, the optimal disassembly depth, and the optimal circular economy strategy. The tool can be used in the development phase, i.e. in the design of the batteries to advance automation with the tool. Rosenberg et al. (Rosenberg et al. 2022) assess the disassembly duration of a battery system under uncertainty applying a fuzzy logic approach. This serves as a basis to derive cost estimations and is supported with actual disassembly experiments. Aguiar et al. (Aguiar et al. 2017) propose a diagnostic tool to evaluate a product's recyclability already during its development phase. The tool makes use of an assortment of EoL indices, and its graphical output makes it possible to sort the disassembly challenges according to their severity. The proposed tool was used in a portable cassette and CD player, simulating its redesign, aiming to improve its end-of-life (EoL) performance.

Despite the growing focus on optimising EoL processes for LIBs several gaps remain in the comparative evaluation and detailed analysis of individual battery component designs to facilitate automated disassembly. Key areas that require attention include:

- There have only been few detailed comparative evaluation of the design of different batteries and none for battery components from different brands to assess the extent to which they facilitate automated disassembly.
- There is a lack of structured design analysis at the component level and comparisons of how they enable or hinder automation of disassembly. Existing publications tend to summarise insights from a higher-level perspective, comparing whole battery packs without going into detail.
- Clear and structured criteria lists have been developed to evaluate automated processing in EoL LIBs disassembly, but these have only been used in exceptional cases to evaluate processes or whole products, and never to evaluate individual component designs.
- There is no study that specifically provides guidance for component design that promotes automated disassembly. Existing guidelines are always rather broad and do not specifically consider individual components

Consequently, Ricard et al. (2024) emphasise the necessity for continued research towards establishing a standardised framework for the design of automated disassembly systems, alongside specific guidelines catering to the design of individual components (Ricard et al. 2024).This literature review highlights the gap in comparative analyses of automated disassembly processes, particularly for EV batteries and their single components. Furthermore, it is important to highlight that the depth of analysis usually ends at the module level and therefore the areas of the periphery and cells are not sufficiently examined (Wu, Kaden, and Dröder 2023). The implementation of automated disassembly for lithium- ion batteries underline the need to investigate all individual battery components and disassembly depths (Xiao, Jiang, and Wang 2023; Zorn et al. 2022).

2. Methodology

Given the anticipated scale of return volumes it is reasonable to include remanufacturer's and recycler's requirements regarding possibilities for automated disassembly in product development. The evaluation methodology developed here thus considers these key stakeholders in product development. The procedure developed for this study that integrates their requirements into battery development is shown in Figure 2.

The procedure consists of four steps that will be explained in the following sub sections. It starts with a disassembly analyses, which is then followed by the examination based on the automated disassembly enabling requirements list. In the third step, a visualisation is employed to identify the highest rated components, from which component specific design guidelines are derived in step 4. The focus of the developed methodology lies on the consideration of the general product design and architecture, as well as the investigation of joints, connectors and fasteners. However, the purposeful DfD-oriented material selection is not a specific focus of the methodology. Ultimately, the application of this methodology enables the comparison of alternative design options for each component, with a subsequent ranking according to performance. The derived design guidelines from the most effective examples can be directly applied in product development.

Step 1: disassembly analyses

To generate a comprehensive overview of the market, battery designs spanning different years of manufacture (2013 – 2021) are selected. Additionally, systems with varying capacities (10 kWh – 71 kWh) from a range of vehicles, including electric vehicles (EV) and plug-in hybrid electric vehicles (PHEV), are considered to ensure a broad and representative analysis. During disassembly analyses all relevant characteristics of each disassembled component from all battery designs are documented. The different manual disassembly experiments are conducted in accordance with the process chain presented in Figure 1 with modifications where necessary to accommodate the design variations of the investigated batteries. In each case the following procedure is applied iteratively:

(1) Identification/analysis of the joints of component to be removed
(2) Unfastening of joints, recording of necessary data for the evaluation
(3) Removing the component (disassemble the component as deeply as possible)
(4) Determination of the mass and dimensions of removed component

The premise for all disassembly steps is to remove each component non-destructively.

Step 2: application of automated disassembly – requirements list on component level

The automated disassembly requirements list used in this study is based on a prior study by Herrmann et al. (Herrmann et al. 2012). The authors devised a list of criteria to develop a scoring model for the evaluation of automation possibilities of processes involved in LIB disassembly. Thus, the methodology is designed for the same general use case. However, as this study is design oriented rather than process

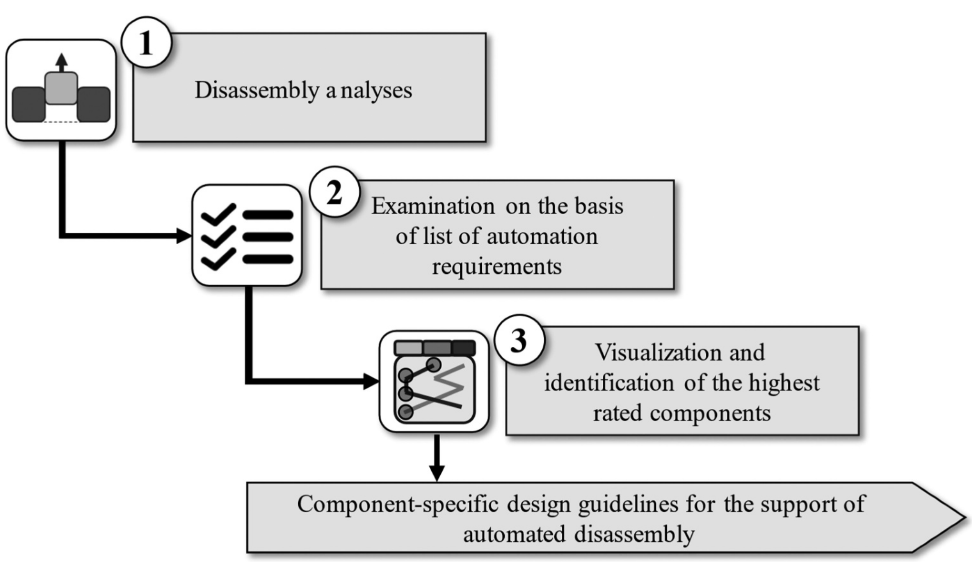


Figure 2. Proposed methodology for deriving component-specific design guidelines for automated disassembly to support requirements elicitation in product development.

oriented it is adapted where appropriate to better reflect the focus of this analysis. Therefore, the existing list of criteria is transformed into a list of requirements that represents the key demands from remanufacturers and recyclers. The complete list of criteria from the original publication can be found in Appendix 2, accompanied with a short description of how each criterion has been dealt with to fit this studies purpose. In general, differences include a deeper focus on the technical possibility of automation in the case of this publication, which is why certain criteria from the original list are left out.

The evaluation system for each requirement is adapted to either one of two evaluation methods. Either a specific value between 0 and 1 is assigned for each possible design characteristic (semi-quantitative; symbol .·˙) or a discrete quantified value is derived calculated by Equation 1 to 5 (Appendix 3: Equations and methods for the assessment of requirements 1, 2, 5, 9 and 10 to generate quantitative results

Table 2 displays the 11 individual requirements, each accompanied by a concise description, the definition of associated rating direction, resulting weighting from pairwise comparison, and the evaluation method. For requirements 3, 4, 5–8 as well as 11 a specific value is given that directly depends on the characteristics of the individual batteries. This is explained in Table 2 in the right column. For requirements 1, 2, 5, 9 and 10 more complex evaluation schemes are applied to derive quantitative results. The equations and methods used to evaluate these requirements is provided in Appendix 3.

The results are visually represented, using a vertical radar chart, where each component's evaluation result is presented individually in horizontal order. This visualisation allows a design comparison and provides a detailed view of the highest evaluated design for each component. The vertical order of component entries reflects the dismantling sequence and allows for the comparative analysis of all different component designs and thus the identification of significant differences. The evaluation results are depicted on a scale ranging from 0 to 1, where 0 indicates that the component is easy to disassemble, and 1 signifies that it is difficult to disassemble. The 'Average' value serves as a benchmark for assessing individual components against all design alternatives. It provides insight into the performance of a specific design relative to the overall average, facilitating a comparative analysis that highlights potential areas for improvement among the components.The value 'Difference Min – Max' elucidates the variance within a component design, highlighting the extent of improvement potential among different designs in relation to each other. Figure 3 schematically represents the general structure of the graph and explains the single elements.

Step 3: Derivation of Design for automated disassembly (DfAD) guidelines

The proposed vertical radar chart designed for this publication and introduced in the previous step shows the best-performing design for each component on the very left. This visual representation helps to identify the superior designs, providing a comprehensive overview for comparison. Based on this analysis, the guidelines are derived by examining these best-performing designs and then considering their unique design characteristics and special features that set them apart. Heemann provides in a study from 2005 a list of criteria that allows for the structured presentation of guidelines (Heemann 2005). The list is given in Appendix 5 and is considered during the application of the methodology proposed here. The criteria concern, for example, the type of summarisation of information and the information structure.

Accordingly, the derivation of design guidelines involves a systematic approach comprising five key steps:

- Identification of the highest rated battery component design
- Examination of the individual requirements that resulted in the highest rated battery component design
- Comparative analysis of the individual component design ratings
- Examination of individual design influences on ratings and derivation of design guidelines
- Assessment based on expert knowledge

This approach ensures that the final list of guidelines not only aligns with the best design practices but is also technically feasible. As the analysis is focusing on market-available battery designs, it is ensured that the recommended solutions have already been proven to be technically viable. This practical constraint helps to bridge the gap between theoretical ideals and practical implementation, making the guidelines achievable. The guidelines are formulated as instructions and in such a way that they can be integrated into the product development process to extend the classical perspective with DfAD criteria.

3. Application and results

To derive meaningful insights into design enabled automation capabilities at component level, the methodology developed here has been applied to different market available battery designs. The following subsections show the results from this analysis.

3.1. Disassembly analyses

In total five battery systems have been disassembled from pack to cell. Details and individual characteristics can be found in Table 3.

The selected batteries represent a wide selection of current state-of-the-art but also former designs. This enabled us to review different generations to draw conclusions on the development of design dependent dismantlability over time but also to include different battery sizes in our considerations. Capacities range from 10 to 71 kWh, thereby representing the current range of typical capacity values for passenger hybrid and full electric vehicles. The investigated designs are exclusively cell – module – pack designs to ensure comparability between them. Cell-to-pack and cell-to-chassis designs are not considered from these considerations.

Table 2. List of requirements to enable automated disassembly by purposeful design. Each requirement is accompanied by a description as well as the and the classification according to the evaluation system (semi-quantitative or quantitative). The table also shows the requirement's relative importance that is derived from the pairwise comparison and the evaluation method used for this study (adapted from (Herrmann et al. 2012).

j	Requirements for automated disassembly		Evaluation quantitative (↗) or semi-quantitative (.⋰) and short description	Weighting [%] *(lj)* rounded	Evaluation method
1	Low number of robot motions	↗	Reduction of the number of robot motions involved to remove the component*	11	Result: 0, . . . , 1 The less movements necessary, the better the evaluation result, Calculation according to appendix equation 1
2	Low number of necessary tools	↗	Reduction of the number of tools necessary to remove the component	10	Result: 0, . . . , 1 The less tools required, the better the evaluation result. Calculation according to appendix equation 2
3	Accessibility for robot arm	.⋰	Availability of space for the robot arm to access and remove the component.	16	1: no access possible; 0.5: access possible, but not from z axis; 0: no access restrictions
4	Joining type separable for robot	.⋰	Evaluation of the joining type and the possibility it provides for the robot to separate, loosen or undo it**	17	0: Reversible joint types; automated disassembly with current tools possible 0.33: Reversible under the right conditions; (e.g. Manually but not yet automated possible) 0.67: Not reversible, destruction of joint type necessary 1: firmly bonded joint type that might also require the destruction of the component if removed/released/disconnected
5	Low accuracy required	↗	Evaluation of the accuracy, that is required for the robot to access the component or the joining	9	Evaluated subtracting the sum of fulfilled requirements (1–0.33 per requirement): • Minimum object size undercut • Sufficient space to operate disassembly tool • Acceptable necessary accuracy to apply the disassembly tool
6	Avoid involvement of hazardous material	.⋰	Evaluation of the potential occurrence of hazardous material, when disassembling the component	9	Yes: 1; No: 0
7	Avoid necessity to disconnect high voltage carrying cables	.⋰	Evaluation of the necessity to interact with current carrying cables disassembling the component	5	0: or cable carrying low voltage 1: cable carrying high voltage
8	Avoid need for customised tools	.⋰	Evaluation of the necessary involvement of customised tools	5	Yes: 1; No: 0
9	Reduce geometrical workpiece dimension	↗	Evaluation of the characteristic length (or longest dimension) of the component that might complicate its disassembly	5	Result: 0, . . . , 1
10	Reduce weight of component to be disassembled	↗	Evaluation of the component's weight that might complicate its disassembly	3	Result: 0, . . . , 1;
11	Ensure acceptable mechanical properties during force application	.⋰	Evaluation whether the component bends under force influence (gravitational or typical force influence during the component's disassembly process)	9	1: Component cannot withstand own weight, bends reversibly or irreversibly 0.67: Component can withstand own weight but might deform plastically during minor/typical force application*** 0.33: Component might deform elastically 0: The component is dimensionally stable/ resistant under the influence of typical force

↗The evaluation value is determined through a quantification by the application of an appendix Equation 1 to 5).
.⋰The evaluation value is determined by a semi-quantitative approach and specific values between 0 and 1 are assigned to specific design characteristics.
*value is scaled to the batteries' capacity.
**if more than one joint-type is used to fasten component, the worse result is considered.
***Typical force application: Force that is required in the process of removal of the considered component.

3.2. Application of automated disassembly-requirements list on component level visualisation and identification of highest rated components

In the following, the results of the evaluation of an exemplary component are discussed in detail to explain the general logic of the results. For this, the system cover is chosen to provide a representative picture of the evaluation, which results from a large number of different design measures. Figure 4 shows the evaluation results for all five investigated designs and the entire list of evaluation criteria. The evaluation results for all further components are integrated in the Appendix 6. The results are colour-coded, with green indicating a high result (good performance) and red indicating a low result (poor performance). Intermediate performance is shown in yellow and orange. Each criterion is scored on a scale from 0 (easy) to 1 (difficult to disassemble), with 0 representing the best possible outcome. The evaluation has been performed by the authors of this publication.

The differences in the results observed for the system cover are mainly attributed to the choice of seal used. All investigated designs employ screws or bolts to fasten the system cover to the pack housing tray to prevent their unintentional opening and to ensure electromagnetic compatibility between the housing components. A conductive joint type should therefore

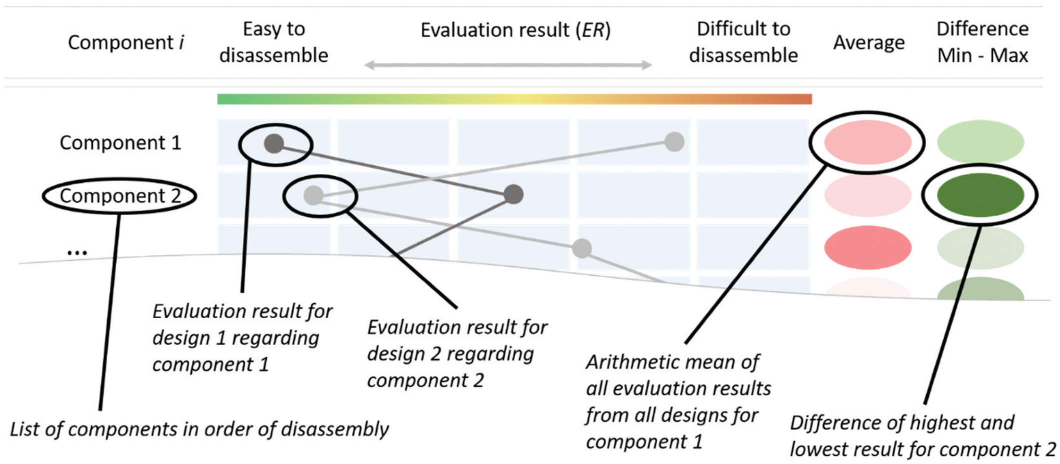


Figure 3. Schematic representation of the vertical radar chart used to visualise the results with the component for which each design alternative is evaluated, their respective evaluation result (*ER*) from easy to disassemble (left) to difficult to disassemble (right). The average represents the arithmetic mean of all evaluation results for the given component. The difference min – Max is the difference between the highest and lowest evaluation result for the given component. The joints between different components are respectively assigned to only one specific component.

Table 3. The battery systems examined in this study and their individual characteristics.

	Unit	BMW	Smart	Mercedes Benz	Audi	Renault
Name	[-]	X1 xDrive25ee	EQ Forfour	eSprinter	e-tron 50 Quattro	Kangoo Z.E.
Vehicle type	[-]	Plug-in Hybrid Electric Vehicle	Electric Vehicle	Electric Vehicle	Electric Vehicle	Electric Vehicle
Battery brand	[-]	Samsung SDI	LG Chem Ltd.	LG Chem Ltd.	Samsung SDI	LG Chem Ltd.
Battery capacity	[kWh]	10	17.6	54	71	22
Module capacity	[kWh]	2	5.66	6.75	2.62	11
Number of modules	[n]	5	3	8	27	2
Year (release on market)	[a]	2020	2017	2018	2015	2011
Year (manufacturing)	[a]	2019	2017	2021	2018	2013
Used cell type	[-]	Prismatic	Pouch	Pouch	Prismatic	Pouch
Cell chemistry	[-]	NMC	NMC	NMC	NMC	NMC
Weight (pack)	[kg]	91.2	160	470	580	230
Weight (module)	[kg]	13	44	43	14	92
Energy density (pack) (calculated)	[kWh/kg]	0.111	0.106	0.114	0.101	0.095
Energy density (module) (calculated)	[kWh/kg]	0.153	0.129	0.157	0.189	0.12

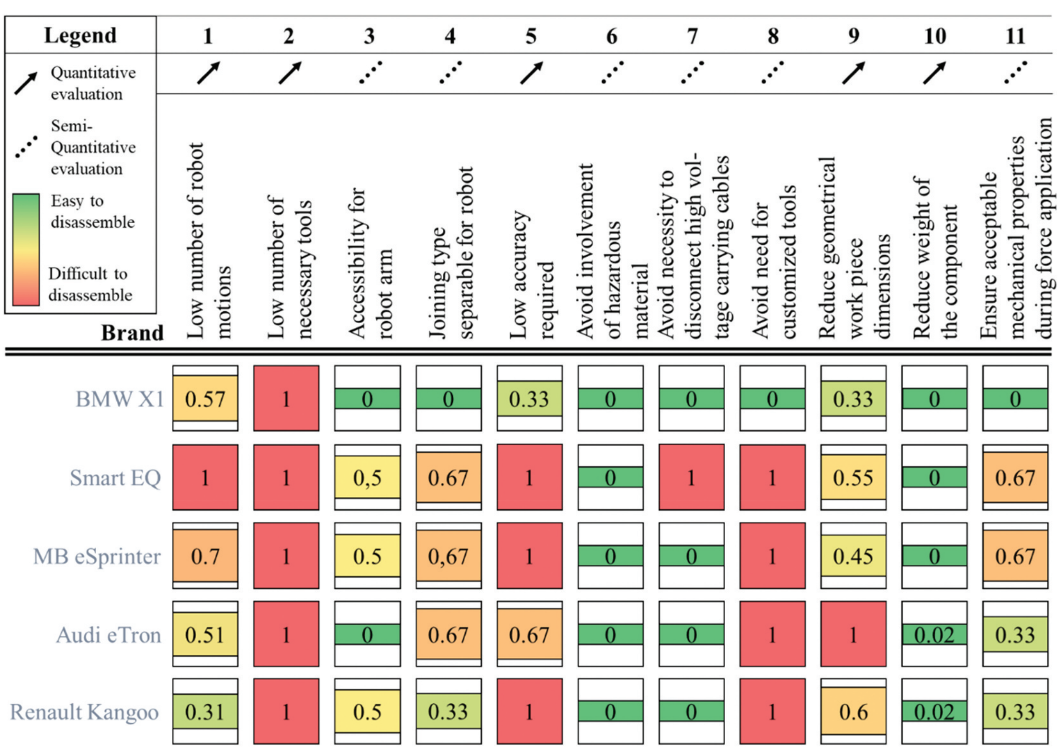


Figure 4. Heatmap of the evaluation results for "1. System cover" component for all 5 investigated designs and the entire list of evaluation criteria; the figure reveals the hotspots, that influenced the results for that component. The weighting of the requirements according to the pairwise comparison is not included here.

prevent other devices from being disturbed by unwanted electrical or electromagnetic effects.

The BMW X1 (2019) system cover was determined to be the easiest to disassemble primarily because it is designed with a seal that is only inlaid and not fixated with adhesives. This allowed for an easy lifting of the cover after unscrewing the bolts. The seal can be easily removed by a robot's gripper in the next step, as it is only secured against displacement by form-fit positioning elements. As a result, the robot arm can easily access the component, and the joining type is separable by the robot without the need for special tools. Additionally, since no force is required to separate the components, no deformation occurs. This results in the evaluation value of 0 for requirements 3, 4, and 11.

All other designs investigated for this study also employ adhesive seals which significantly influences the bonding between the system cover and the base of the housing, thereby increasing the complexity involved during the lifting of the system cover. For the investigated designs often a high accuracy of the cutting tool was necessary to hit the gap between the cover and tray, resulting in high values for requirements 3, 4 and 5. Further, it was necessary to apply force to overcome the high adhesive forces which often resulted in a permanent deformation of the system cover, especially for the Smart EQ and MB eSprinter leading to high values for requirement 11. The system cover of the Smart EQ is designed in such a way that current carrying components must be detached (requirement 7) before it can be removed.

In this subsection the visualisation of the calculations for all battery systems and components is shown. It displays the average score and the difference between the worst and best value of the individual components and forms the basis for the to derive the DfAD guidelines. Figure 5 shows the evaluation results of the disassembled battery system designs, that are plotted as a vertical radar chart.

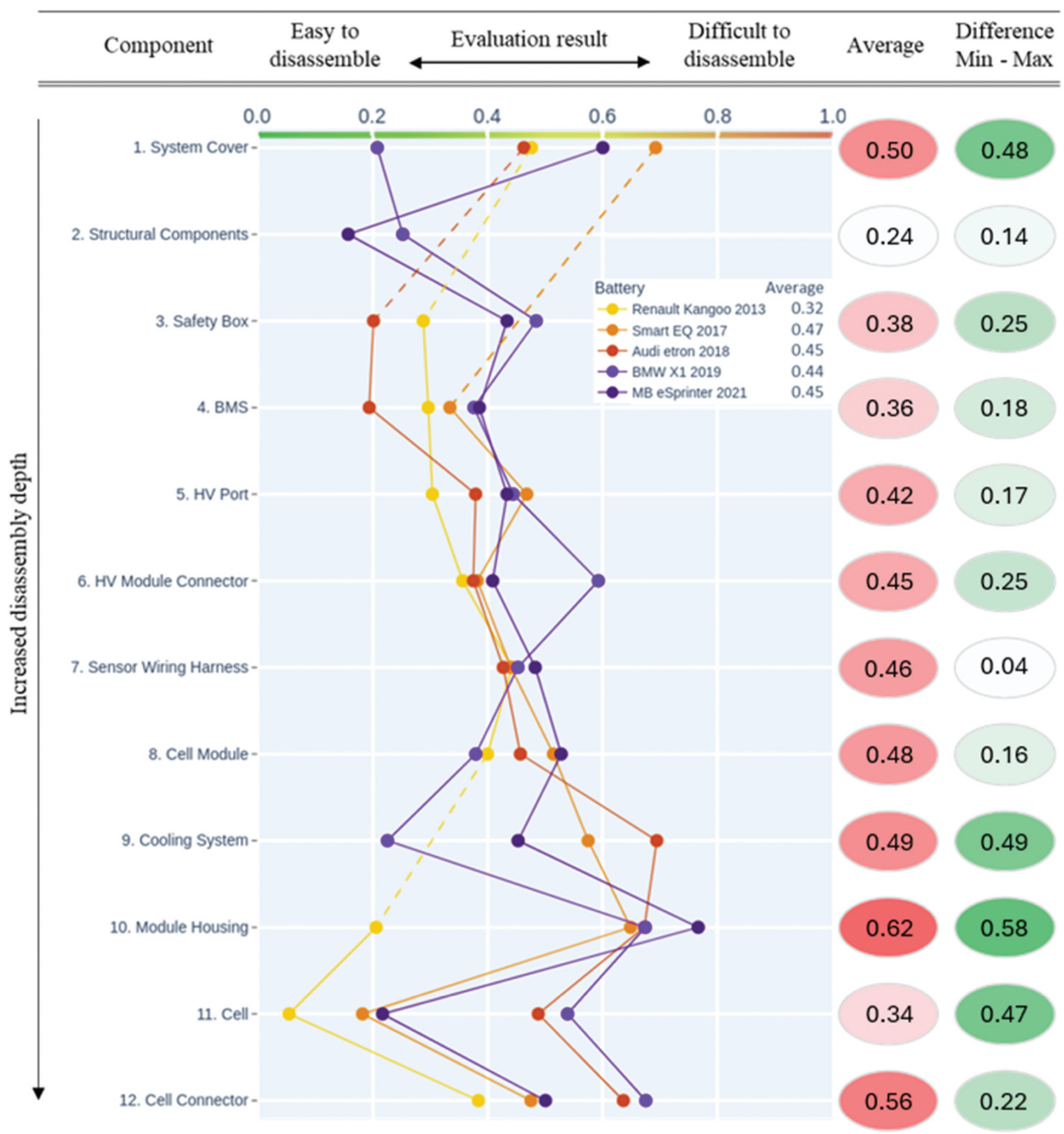


Figure 5. Evaluation result of the investigated designs for all components. The results were received by the application of Equation 1 and thus represent the weighted average of all applied evaluation criteria. The depiction can be considered a vertical radar chart that can also visualize the disassembly direction and reveals further information in form of heatmaps on the Figure's right side. The further left an entry is, the better the design's performance regarding the developed requirements list. The line colours also indicate the age of the batteries from the oldest examined system in light yellow to the most recent in dark purple.

On the vertical axis the 12 different component types that were subject to disassembly are listed in a subsequent order as also displayed in Figure 1. The order thus represents an increased depth of disassembly with continued progression downwards. The horizontal axis shows the design and component specific evaluation results across all 12 evaluation categories, that were calculated according to Equation 1 The further left the entry on the horizontal axis, the better the evaluation result of the specific design and the more the specific design fulfils the requirements listed in Table 2. The arithmetic mean of the design assessment results for each component is shown on the right-hand side of the diagram, as is the difference between the smallest and the largest assessment result.

The allocation of a specific design's component to the entries on the vertical axis was done in a function-oriented manner, as the realisations can differ between the designs. This means that the specific designations must be rather broad. To elaborate, a cell connector can for example indicate that either a busbar or cables are used for module intern cell circuitry. The function-oriented allocation also means, that the heat-conducting paste applied in most designs would be allocated to the cooling system. This is independent from where it is applied and in which specific disassembly step it would have to be removed. This allocation was necessary to ensure a proper basis for the comparison. Some pack designs do not incorporate specific components as certain functions can be realised in different ways. This is indicated when the plot shows no entry for a component in form of a dot for a design and directly connects the closest entries above and below. Based on the investigated batteries, it appears that the further the dismantling procedure has progressed, the higher the evaluation results get, which indicates that the disassembly becomes more difficult. This is also shown by the 'Average' values of the individual components of the five different designs. This result can be justified by the more intensive use of adhesives and welding joint types at the module levels of disassembly. However, this trend is countered by outliers (see following sub-section). Figure 5 further reveals that many of the components with high evaluation results also exhibit a larger deviation among the investigated designs. This is represented by the difference between the lowest and highest value of the evaluation results obtained. The scatter is quite wide for the first and the four last disassembly steps, while the mid-level disassembly steps show rather similar results for all designs. Components 8 (cell module) and 9 (cooling system) warrant particular attention due to their nearly identical average values of 0.48 and 0.49, respectively. While these values suggest a close performance level, a significant divergence is observed when evaluating the 'Difference Min – Max' metric. This disparity underscores the substantial variability within the assessed designs, indicating that component 9 (cooling system) harbours considerable potential for design enhancements. With design choices which have already been realised and are implemented in series production vehicles.

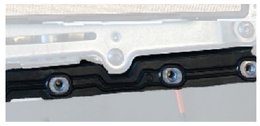
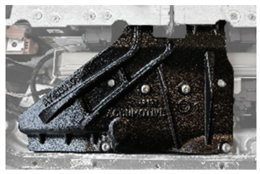
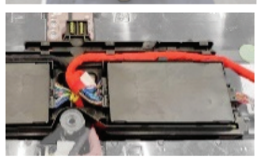
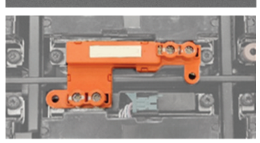
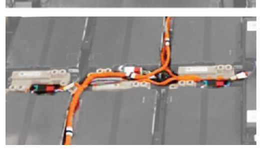
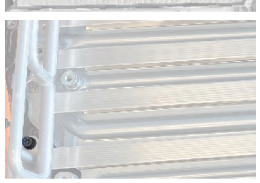
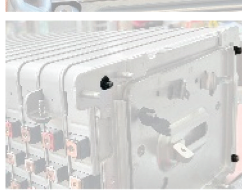
Figure 5 also shows the average values of disassembly difficulty associated with the five investigated designs as well as the year they have been introduced to the market. Notably, the battery pack allowing for the easiest disassembly process is also by far the earliest design, while the remaining more current designs seem to exhibit similar difficulties with a rather increasing trend.

The results for four component specific evaluations revealed that their disassembly is particularly difficult with the highest average evaluation results. Apart from the system cover that has already been described in the previous sub section, these are the cooling system, the module housing and the electrical cell connectors. The cooling system, housing and the electrical cell connectors will be presented in the following.

The analysis of the cooling systems revealed notable variations among the investigated designs. The design specifications of BMW X1 entailed the best evaluation results due to its easily detachable cooling system that can be removed by loosening a small number of bolts. The cooling systems of the Smart EQ and MB eSprinter are quite similar in design, which resulted in similar evaluation result. Both are affixed to the base of the battery systems with a combination of screwing and a thermally conductive adhesive (see for example Figure 6; picture 2). The cooling

Figure 6. Pictures taken during the disassembly trials, showing special characteristics of different designs 1: BMW X1: opened welding seams of a module, 2: SMART EQ: cooling system glued to the system tray, 3: MB eSprinter: modules encased in a metal cage with welding seams and rivets, 4: Renault Kangoo: module housing with lined up cells, 5: MB eSprinter: side view of a module with soldered sensors and bolted cells.

Table 4. Component-specific design guidelines derived from the best-performing battery designs. The guidelines have been derived by closely examining these designs and results from their description also in comparison to worse performing designs.

#	Component	Highest rated design	Depiction of highest rated design	Derived design guidelines
1.	System Cover	Adhesive-free insert seal		• Use an adhesive-free seal • Ensure bending resistance • Through purposeful material selection and/or sufficient material thickness • Reduce number of bolts
2.	Structural Component	Design with few bolts		• Design the battery housing to be stiffer to make structural components obsolete • Ensure easy access to bolts
3.	Safety Box	Combined plugs		• Ensure easy access to bolts • Ensure easy access to plugs • Enable easy opening of the plugs with a robotic tool • Reduce number of actions by combining plugs
4.	BMS	Combined plugs		• Bundle plugs to reduce the quantity • Ensure easy access to bolts • Ensure easy access to plugs • Enable easy opening of the plugs with a robotic tool • Reduce number of actions by combining plugs
5.	HV Port	Design with good accessibility		• Use an adhesive-free seal • Ensure easy access to bolts
6.	HV Module Connector	Bolted busbars		• Ensure easy access to bolts and busbars • Ensure bending resistance through sufficient material thickness
7.	Sensor Wiring Harness	Combined plugs		• Bundle plugs to reduce the quantity • Ensure easy access to plugs • Enable easy opening of the plugs with a robotic tool
8.	Cell Module	Adhesive-free module mounting		• Use reversible joint types • When applying adhesive: • Enable a handling point for extracting the module • Enable the module to be removed with minimal force
9.	Cooling System	Bolted cooling system		• Use reversible joint types • When applying adhesive: Ensure at least 1 mm of vertical space at the edge (e.g. via rounded groove/bevel) to allow for the insertion of a cutting tool • Ensure bending resistance • Through purposeful material selection and/or sufficient material thickness
10.	Module Housing	Bolted housing components		• Use reversible joint types • Ensure early access to individual cells (without having to disassemble the entire module first) • When using rivets and/or welding seams: Ensure access to allow for their save breaking

(*Continued*)

Table 4. (Continued).

#	Component	Highest rated design	Depiction of highest rated design	Derived design guidelines
11.	Cell	Adhesive-free cell alignment		• Use an adhesive-free alignment • When applying adhesive: Ensure at least 1 mm of space at the edge (e.g. via rounded groove/bevel) to allow for the insertion of a cutting tool • Ensure electrical insulation for the safe usage of robotic tools
12.	Electric cell connection	Bolted electric connections		• Use reversible joint types • When applying welded seams: ensure at least 1 mm of vertical space at the edge (e.g. via rounded groove/bevel) to allow for the protection of the cells during disassembly • Add non-conductive material as pole-covering to avoid short circuits during disassembly • Allow for horizontal access to the cell connections

system design of the Audi e-tron is both difficult to detach and hard to access. It is permanently installed between the module tray and the underride guard, which are welded together. To ensure the cooling system functions properly, a thermally conductive adhesive is also integrated between the module and the tray which cannot be disassembled. The Renault Kangoo's module housing design was identified to be the easiest-to-open. This results from the avoidance of material bonding joint types, and the arrangement of cells in a linear layout that are fastened with end plates and threaded rods. In contrast, the BMW X1 and Audi e-tron designs feature cells enclosed between tension anchors with end plates that are joined by welded seams, as shown in Figure 6; picture 1. As a result, the disassembly of these battery modules requires the breaking up of firmly bonded joint types. To dismantle the modules of the Smart EQ and the MB eSprinter, it was first necessary to unscrew bolts fastening the cells, that are individually connected to the surrounding metal cage. The metal cage was opened by abrasively cutting and breaking the rivets that seal the cage. As the rivets are more durable than the surrounding material, separation required a high level of accuracy to avoid damages to the surrounding parts or heating up the cells. The module housings of the MB eSprinter share central design elements with the Smart EQ, with the additional disassembly obstacle that the metal cage is sealed with welded seams, in addition to the rivets (see for example Figure 6; picture 3 and 5).

Regarding the electrical cell connections (Figure 6; picture 1), it was identified that the design employed for the Renault Kangoo yielded the best result. This is due to its bolted cell connections. In contrast, the Smart EQ and MB eSprinter utilise spot welds to connect neighbouring cell tabs, which can be easily yet only irreversibly separated with an insulated cutting tool. Separating the prismatic cells of the BMW X1 and Audi e-tron required the greatest effort. Their cells are joined together by welded aluminium sheets, which must be carefully chipped off individually with a chisel-like tool

3.3. *Derivation of design for automated disassembly guidelines*

As mentioned in Section 1, the aim is to seamlessly integrate the results of this publication into existing product development procedures. A suitable starting point for the disassembly-oriented design development is to combine the components with the best evaluation results of Figure 5 into a new product, similar to the procedure of a morphological box. Therefore, the design guidelines are derived from the best-performing design examples. Table 4 contains the entire list of design guidelines derived for each component.

The guidelines are formulated in such a way that they meet the previously defined presentation criteria (Section 0, step 4 and Appendix 5). Accordingly, they are kept concise and formulated as commands and accompanied by a short depiction of the best performing design. These pictures highlight the design characteristic that contributed most to the result. The design guidelines derived primarily focus on the exclusion of adhesives in the batteries' design, which has been pointed out before. The novelty of this analysis, however, is that the technical feasibility of the requirements is guaranteed since only existing battery designs have been investigated. A common challenge with proposals for design adaptations that aim to increase a product's circularity is to guarantee their technical feasibility. Since for this study only already existing and market available designs are investigated, this is not an issue, and the technical feasibility can be considered proven. Furthermore, the accessibility of the interfaces and the use of reversible joint types play a decisive role in enabling automated disassembly processes.

4. Conclusions and limitations

Current battery designs available on the market are characterised by a high degree of heterogeneity. This is justifiable, as electric mobility is a comparatively new technology that is developing rapidly on the market. Many technological challenges still need to be overcome and there is potential for improvement in many areas, such as the increase in energy density and possible charging velocity. In current product development, these requirements are often given priority over the requirements of a disassembly-oriented design, which results in a high effort during disassembly and prevents its automation.

In the framework of this publication detailed disassembly experiments of five market available battery designs have been conducted. These designs have been evaluated on component level by the application of a requirements list to identify the component designs that best promote automated disassembly. The knowledge gained in this way has been incorporated into component specific design guidelines.

The evaluation led to the identification of focus components that most hinder efficient dismantling of the battery system, namely the system cover, the cooling system, the cell module housing and the electrical cell connectors. Based solely on the findings presented here, it could be inferred that design-dependent disassembly challenges have increased over time, particularly from the level of individual modules downward. Results show that automated disassembly is promoted by the provision of access to the connections and the avoidance of material bonded joint types, which has been identified as disassembly-promoting measure before. Nevertheless, the guidelines proposed in this study are based on designs that have already been proven to be technically feasible and could thus be directly applied as guidance for future battery design, possibly in the framework of the already established V-model for product development.

The methodology developed here considers the dismantlability of a design with different characteristics and disregards other requirements. It is therefore improbable that simply selecting the best performing individual designs identified for each component in this work would result in a product that could satisfy the various demands that the battery must fulfil. Especially, as the design requirements for battery systems seem to have become increasingly challenging in the last years, it is necessary to strike a compromise and find accordingly feasible design solutions and to solve conflicting goals.

The criteria catalogue from Herrmann et al., used to develop disassembly-enabling design requirements, including the need for process automation. Although the catalogue considered hazardous materials, none were encountered in this study, so the criterion had no influence but was retained for future applicability. As in the original study, requirements were weighted through pairwise comparison in a workshop setting. While the selected batteries vary in design, OEM, and capacity, it cannot be considered representative of the current market situation. The case study focuses on existing designs, thereby limiting the identified measures to a retrospective perspective. Recent innovations, such as cell-to-x architectures, are not included in the investigation.

The results must be analysed considering the fact that the population of five dismantled battery systems is too small to guarantee the general validity of the findings. It can be assumed that the diversity of battery designs available to some extent limits the significance of the results, meaning that they only apply to the battery systems analysed here. However, trends can be derived to a certain extent.

Overall, the methodology applied in this study proved to be able to provide a good basis to compare different designs. The use of the method enables OEMs to benchmark their own current batteries against others. On the one hand, this allows the company's own design features to be benchmarked against other manufacturers, and on the other hand, trends across different battery generations and vehicle segments can be identified. The results of the assessments conducted for this publication show a diverse range of influences on the potential for automated disassembly, as observed across different component designs. In general, it is shown that the more recent the battery system is, the more difficult it seems to be to dismantle. However, this trend has to be confirmed by analysing further battery systems. The specific presentation of the data allowed the identification of hotspots regarding disassembly difficulty but also revealed the designs that would be qualified to lower the difficulties. The derived design guidelines could be included in product development processes, for example in the framework of the V-model.

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All have read and approved the final version of the manuscript.

Data availability statement

The data that support the findings of this study are available from the corresponding author, G.O., upon reasonable request.

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